



CUBAN NATIONAL CHEMISTRY CONTEST 2018

“The future of our country must necessarily be a future of men of sciences, must be a future of men of thinking, because that is precisely what we are creating...”

Fidel Castro

Name(s): _____ Last name(s): _____

School: _____

Municipality: _____ Province: _____

CODE: _____ Test: **DAY 1. COMMON EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 1 problem set, with a common exam for the three grades. The exam has 4 problems (problem 1, 20 questions; problem 2, 2 questions; problem 3, 4 questions; problem 4, 2 questions) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

												13 3A		14 4A	15 5A	16 6A	17 7A	18 8A
1 H 1.008	2 2A																	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18	
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95	
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)	
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)	

Problem 1. Miscellanea (20 points)

1.1	...	Total	Total
1	idem	20	20

Select the correct answer in each case.

1.1. When the following substances are ordered according to growing lattice energy, which is the correct order? MgO, NaF, CaS

- A) $\text{MgO} < \text{NaF} < \text{CaS}$ B) $\text{CaS} < \text{NaF} < \text{MgO}$ C) $\text{NaF} < \text{CaS} < \text{MgO}$ D) $\text{CaS} < \text{MgO} < \text{NaF}$

1.2. Which of the following oxides is the best reductant reagent? CO_2 , NO_2 , SiO_2 , SO_2

- A) NO_2 B) SiO_2 C) CO_2 D) SO_2

1.3. Among the molecular substances S_8 , H_2O , NH_3 , P_4 , ¿which has the highest melting point?

- A) NH_3 B) H_2O C) S_8 D) P_4

1.4. The combustion of methylamine, CH_3NH_2 , originates CO_2 , N_2 y H_2O . ¿What amount of O_2 is required for the complete reaction of 1,0 mol of methylamine?

- A) 2,5 B) 2,25 C) 4,5 D) 3,0

1.5. The reaction $2\text{NO}(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g}) + \text{N}_2(\text{g})$ follows the rate law $v = k[\text{NO}]^2[\text{H}_2]$. Which are the units of k if the time is given in seconds and the concentration in mole per liter?

- A) $\text{L} \cdot \text{s} \cdot \text{mol}^{-1}$ B) $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$ C) $\text{L}^2 \cdot \text{mol}^{-2} \cdot \text{s}^{-1}$ D) $\text{mol}^2 \cdot \text{L}^{-2} \cdot \text{s}^{-1}$

1.6. A manganese oxide contains 30,4% mass of O. Which is the empirical formula?

- A) MnO_2 B) Mn_3O_4 C) Mn_2O_3 D) MnO

1.7. A temperature rise generally increases the reaction rate in general. Which of the following properties allows us to explain the previous fact?

- A) pressure B) concentration C) energy of collision D) contact surface

1.8. How many unpaired electrons have a chromium atom?

- A) 3 B) 4 C) 5 D) 6

1.9. Which atom has the highest second ionization energy?

- A) Al B) Mg C) K D) Na

1.10. ¿Which periodic properties diminish in group 1 of the periodic table when increasing atomic number? 1. electronegativity, 2. atomic radius, 3. electron affinity

- A) only 1 B) 1 and 2 C) 1 and 3 D) 2 and 3

1.11. How many isomers have the molecular formula $\text{C}_4\text{H}_{10}\text{O}$?

- A) 4 B) 5 C) 6 D) 7

1.12. What is the molecular geometry of the azide ion, N_3^- ?

- A) angular B) lineal C) T-shape D) triangular

1.13. Which of the following species is polar? 1. BrF_3 , 2. CF_4 , 3. SbF_5 , 4. SF_6

- A) only 1 B) only 3 C) 1 and 3 D) 3 and 4

1.14. In an exothermic reaction is valid that

- A) $Q = 0$ B) $\Delta T < 0$ C) $\Delta H < 0$ D) $\Delta E < 0$

1.15. When the following species are ordered according to increasing N–O bond distance, which is the correct order? NO_2^- , NO_2^+ , NO_3^- .

- A) $\text{NO}_2^- < \text{NO}_2^+ < \text{NO}_3^-$ B) $\text{NO}_3^- < \text{NO}_2^- < \text{NO}_2^+$ C) $\text{NO}_2^+ < \text{NO}_3^- < \text{NO}_2^-$ D) $\text{NO}_2^+ < \text{NO}_2^- < \text{NO}_3^-$

1.16. What is the hybridization of the carbon atoms in propyne?

- A) sp^2 B) sp^3 C) sp , sp^3 D) sp , sp^2 , sp^3

1.17. When metals Mg, Zn and Pb are ordered according to growing reactivity with non-oxidizing acids, which is the correct order?

- A) $\text{Mg} < \text{Zn} < \text{Pb}$ B) $\text{Pb} < \text{Zn} < \text{Mg}$ C) $\text{Pb} < \text{Mg} < \text{Zn}$ D) $\text{Mg} < \text{Pb} < \text{Zn}$

1.18. Which of the following acids has a notable acid character? 1. CrO_3 , 2. MnO_2 , 3. SO_2 , 4. P_2O_3

- A) 2, 3, 4 B) 1, 2, 3 C) 1, 2, 4 D) 1, 3, 4

1.19. Which of the following binary substances doesn't have a chemical bonding with strong ionic character? 1. CsAu , 2. MgO , 3. GaAs , 4. CaC_2

- A) 1 B) 2 C) 3 D) 4

1.20. Which of the following metals doesn't form an oxide when burnt in the presence of dioxygen? 1. K, 2. Be, 3. Ca, 4. Li

- A) 1 B) 2 C) 3 D) 4

Problem 2. Identification of substances (10 points)

2.1	2.2	Total	Total
8	12	20	10

A student received the assignment to identify the substances constituting two solid samples (mixture A and B). It is known that sample A may contain SiO_2 , Na, NiCl_2 , CrCl_3 , CoCl_2 or $\text{Ba}(\text{NO}_3)_2$; while sample B may be formed by AgNO_3 , Ag, $\text{Cu}(\text{NO}_3)_2$, KI or Fe. The student reacted each of the samples with common inorganic reagents and obtained the following results:

Mixture A: It is green. It dissolves completely in water originating a green solution and there is no observable evidence of any strong exothermic reaction. After the addition of sodium hydroxide to the resultant solution, a green precipitate is obtained which partially dissolves in excess of the reagent forming an intensely colored green solution. The solid portion is dissolved when adding hydrochloric acid. The addition of sulfuric acid to the original solution doesn't form any precipitate.

Mixture B: It is grey. It dissolves partially in water originating a colorless solution. After several minutes of standing, the solid is inert to hydrochloric acid or sodium hydroxide. On the contrary, when the colorless solution is treated with sodium hydroxide, a white precipitate forms, which is completely soluble in excess of the reagent.

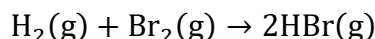
2.1. Find the substances composing mixtures A and B.

2.2. Write the equations corresponding to the chemical reactions that happened during the identification.

Problem 3. Kinetics (10 points)

3.1	3.2	3.3	3.4	Total	Total
5	5	4	6	20	10

From a kinetic point of view, many chemical reactions can be classified as chain reactions. In chain reactions, an intermediate formed in one step generates another intermediate in the following step and so on. These intermediates are named propagators. An example of such a process is the reaction between hydrogen and bromine.



The following mechanism has been proposed to explain the results of the kinetics studies.

Initiation	$\text{Br}_2 \rightarrow 2\text{Br}$	$v = k_1[\text{Br}_2]$
Propagation	$\text{Br} + \text{H}_2 \rightarrow \text{HBr} + \text{H}$	$v = k_2[\text{Br}][\text{H}_2]$
	$\text{H} + \text{Br}_2 \rightarrow \text{HBr} + \text{Br}$	$v = k_3[\text{H}][\text{Br}_2]$
Retardation	$\text{H} + \text{HBr} \rightarrow \text{H}_2 + \text{Br}$	$v = k_4[\text{H}][\text{HBr}]$
Termination	$2\text{Br} \rightarrow \text{Br}_2$	$v = k_5[\text{Br}]^2$

3.1. Identify reactants, products and intermediates. State the molecularity of the last step of the reaction. Which is the rate-limiting step?

3.2. Write down mathematical expressions relating the variation of the concentration of the intermediates in time as a function of the rate constants of individual steps and the concentrations of the species involved.

The experimental rate law for the previous reaction is a complex expression (see below); however, under certain experimental conditions, the expression is simplified and a pseudo-order behavior is observed.

$$\text{Rate law: } \frac{d[\text{HBr}]}{dt} = \frac{k[\text{H}_2][\text{Br}_2]^{3/2}}{[\text{Br}_2] + k'[\text{HBr}]}$$

3.3. Which are the partial reaction orders with respect to Br_2 at the exact moment the reaction starts and when it is close to its end? Justify your answer.

3.4. Find mathematical expressions for the total activation energy and pre-exponential factor, assuming that the reaction rate is $k = 2k_2(k_1/k_5)^{1/2}$.

Problem 4. Stoichiometry I (10 points)

4.1	4.2	Total	Total
4	16	20	10

A sample containing copper, zinc and aluminum of mass 3,000 g reacted with an excess of hydrochloric acid and 1,643 L (measured at 25,00 °C and 101,3 kPa) of gas, a solution and a solid were obtained. The solid residue was filtered, dried and weighed, having a mass equal to 0,850 g.

4.1. Write the equations corresponding to the chemical reactions that happened.

4.2. Find the composition (in mass percent) of the initial sample.